

Fig. 2: An actual-size, single-side PCB for battery tester

it in a suitable plastic case. Use two crocodile clips for connecting battery terminals firmly. Fix potmeter VR1 such that it is accessible for calibration whenever required. There is no need of external power supply.

For calibration, apply 6V-7.5V to the input (BUT) and adjust preset VR1 until LED2 completely turns off. Connect the battery to be tested with suitable crocodile clips and wait for a few seconds to let capacitor C8 reach

its full charge capacity. LED1 should illuminate at a constant intensity, independent of battery voltage. If LED1 illumination is very weak in brightness, or it is completely off, it indicates that the battery is of no use.

If LED1 illuminates well, press S1 and keep an eye on LED2. If LED2 remains completely off, it shows that the battery is in a very good state. If LED2 illuminates brightly for a few seconds, it shows that the battery is weak. This

condition is confirmed by a noticeable weakening in the brightness of LED1. If LED2 illuminates weakly for a few seconds but LED1 illuminates at the same intensity of light, it indicates that the battery is still good but is not new.

After connecting the battery, first verify the test points' voltage levels given in the table. ●

*The author is an electronics hobbyist*

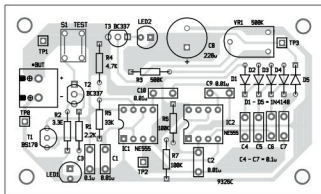


Fig. 3: Component layout for the PCB



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